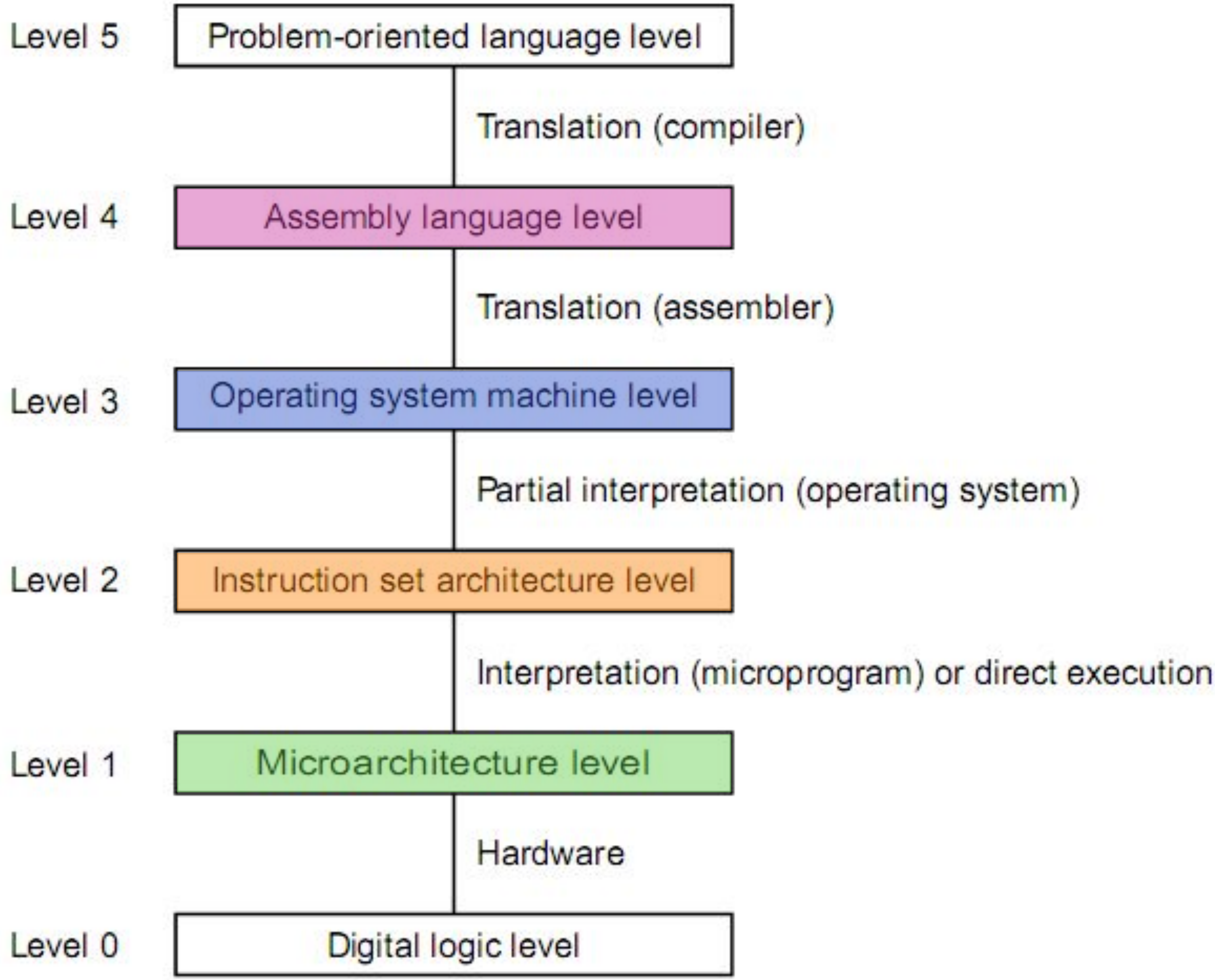
Máquina Multinivel

Componentes de una arquitectura

Arquitectura de computadoras

Pablo A. Montini
Juan I. Iturriaga
Franco Lanzillotta


Level 5
Problem-oriented language level
Translation (compiler)
Level 4
Assembly language level
Translation (assembler)
Level 3
Operating system machine level
Partial interpretation (operating system)
Level 2
Instruction set architecture level
Interpretation (microprogram) or direct execution
Level 1
Microarchitecture level
Hardware
Level 0
Digital logic level

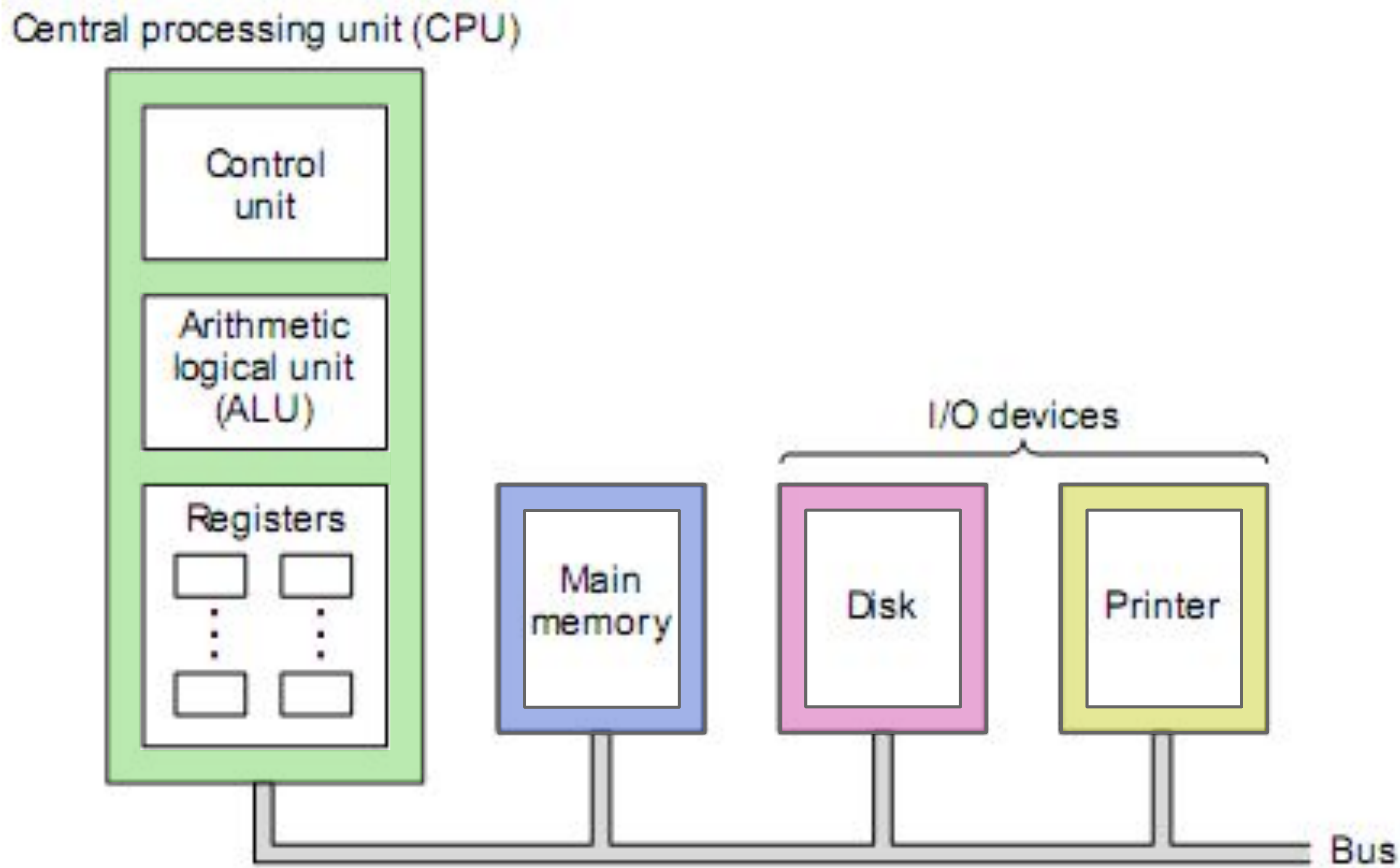


Figure 2-1. The organization of a simple computer with one CPU and two I/O devices.

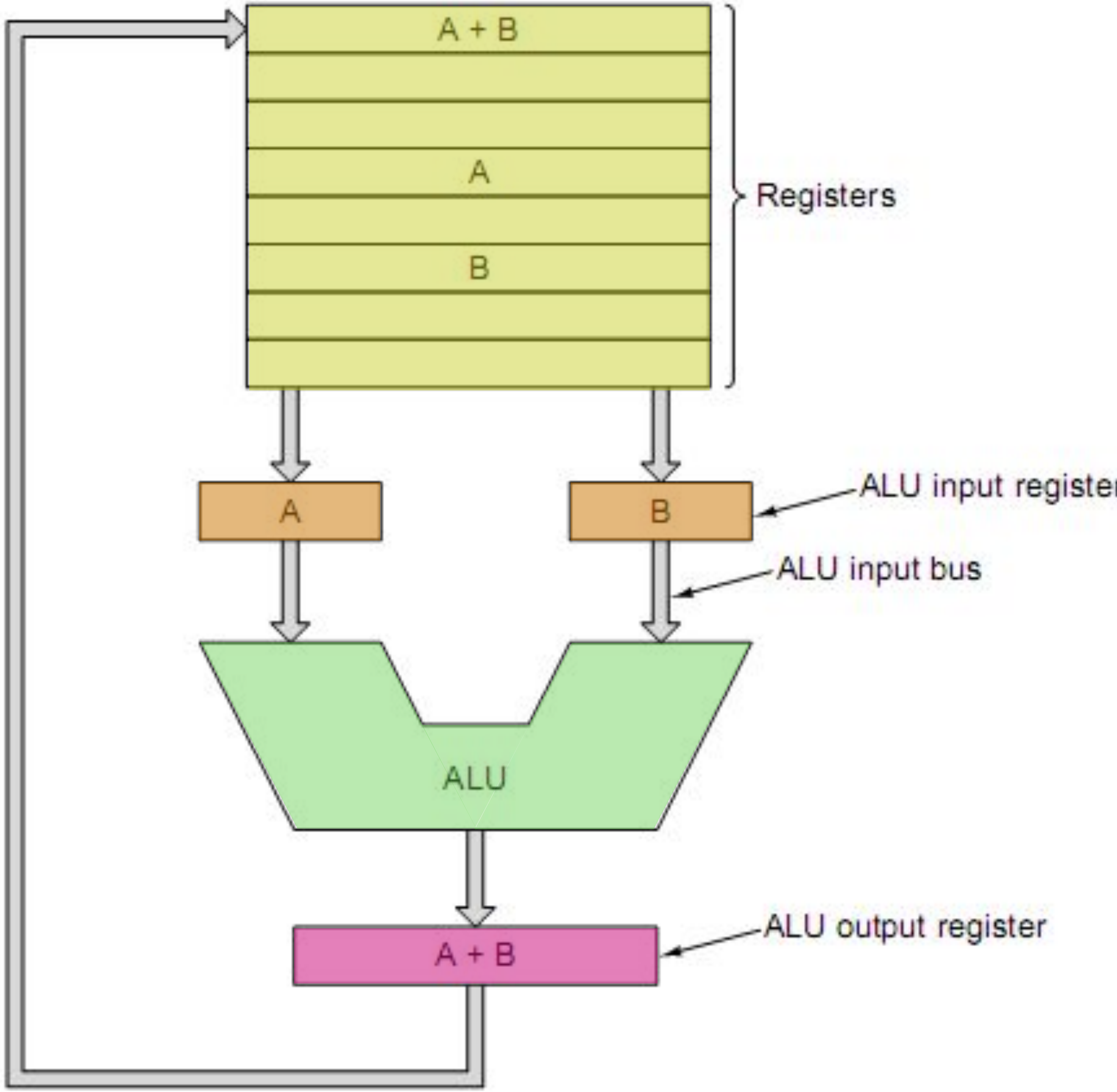


Figure 2-2. The data path of a typical von Neumann machine.

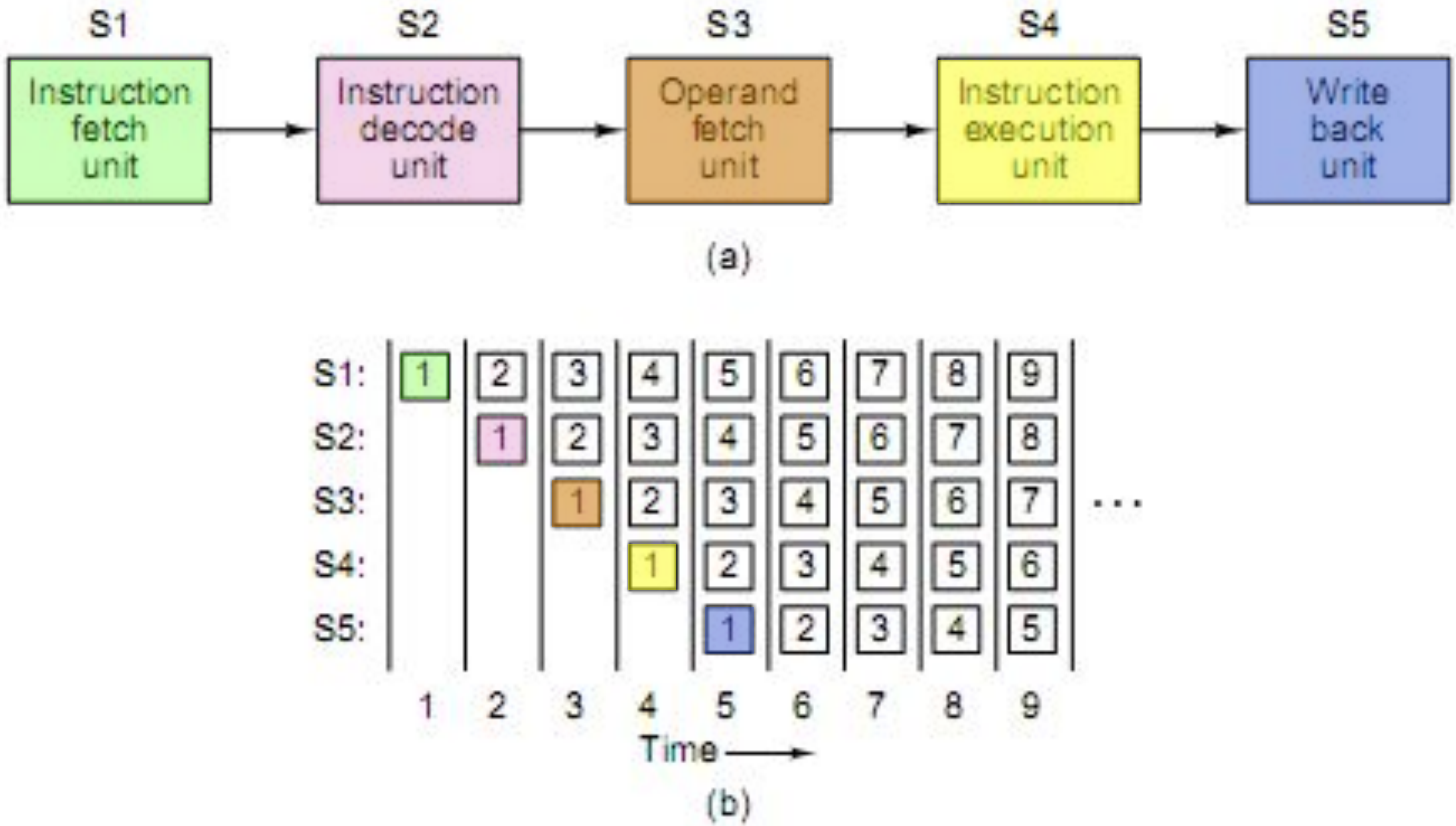


Figure 2-4. (a) A five-stage pipeline. (b) The state of each stage as a function of time. Nine clock cycles are illustrated.

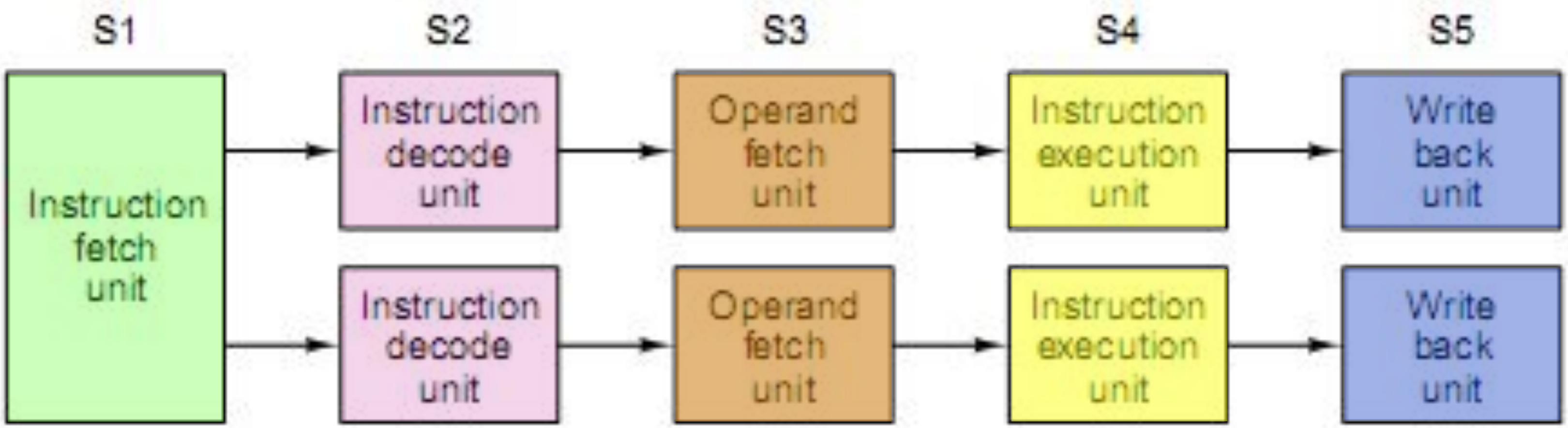


Figure 2-5. (a) Dual five-stage pipelines with a common instruction fetch unit.

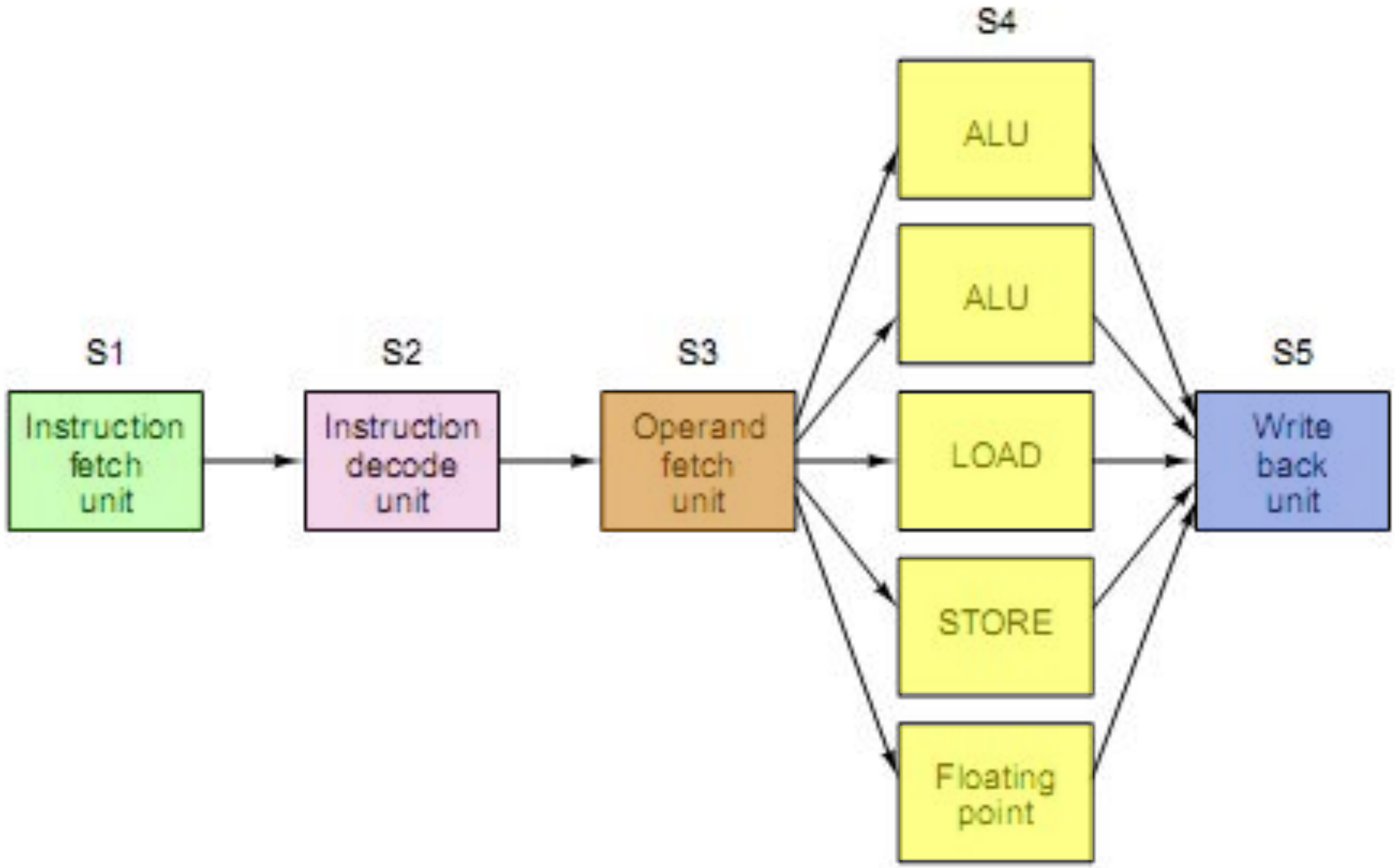


Figure 2-6. A superscalar processor with five functional units.

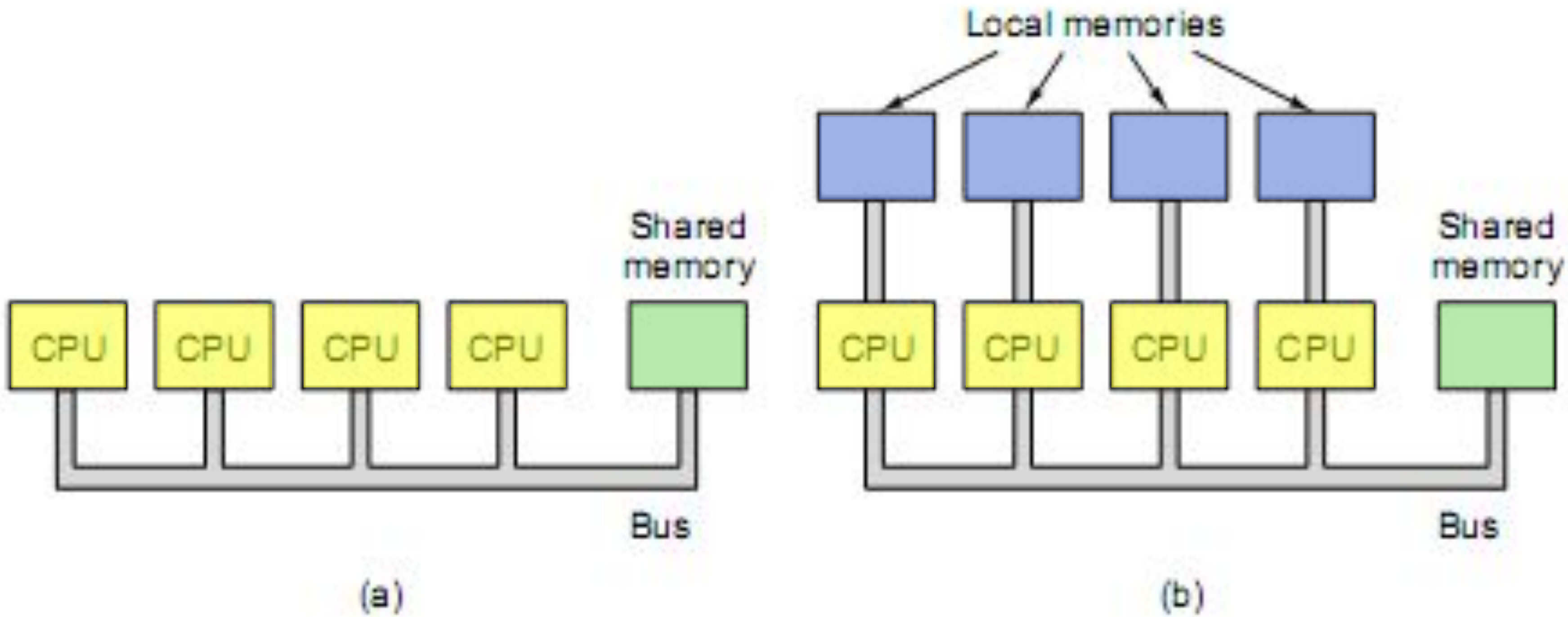


Figure 2-8. (a) A single-bus multiprocessor. (b) A multicomputer with local memories.

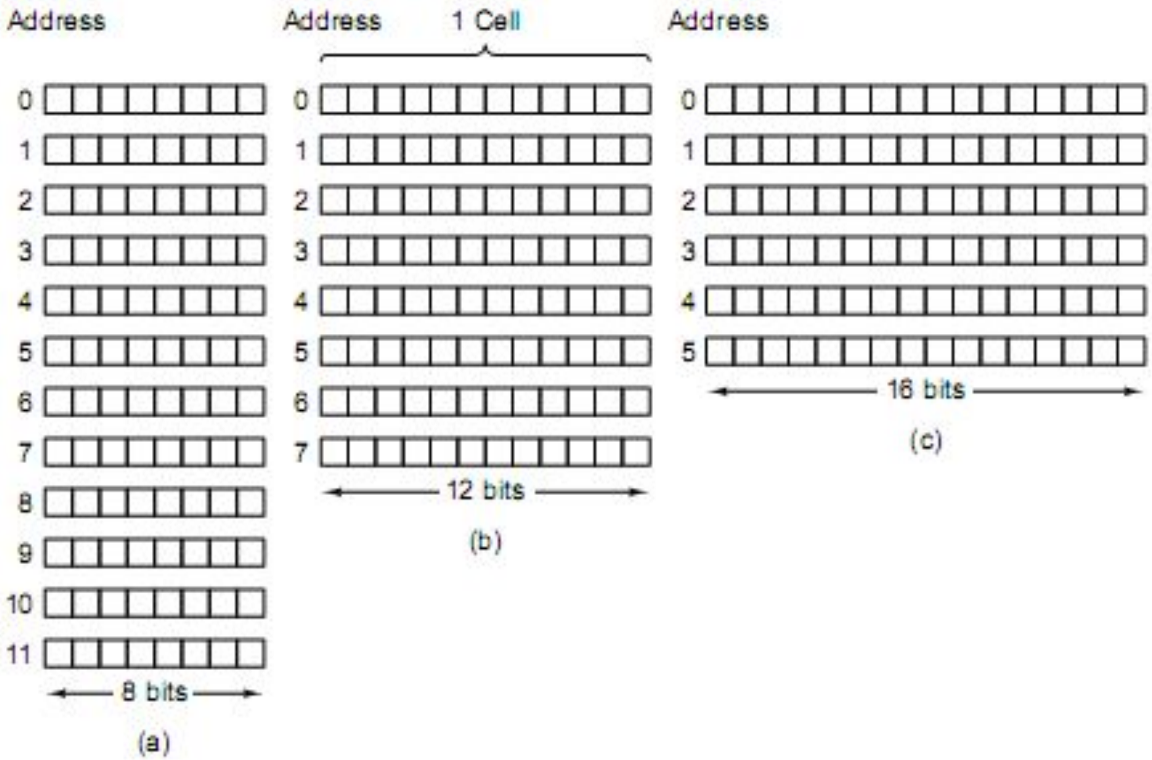
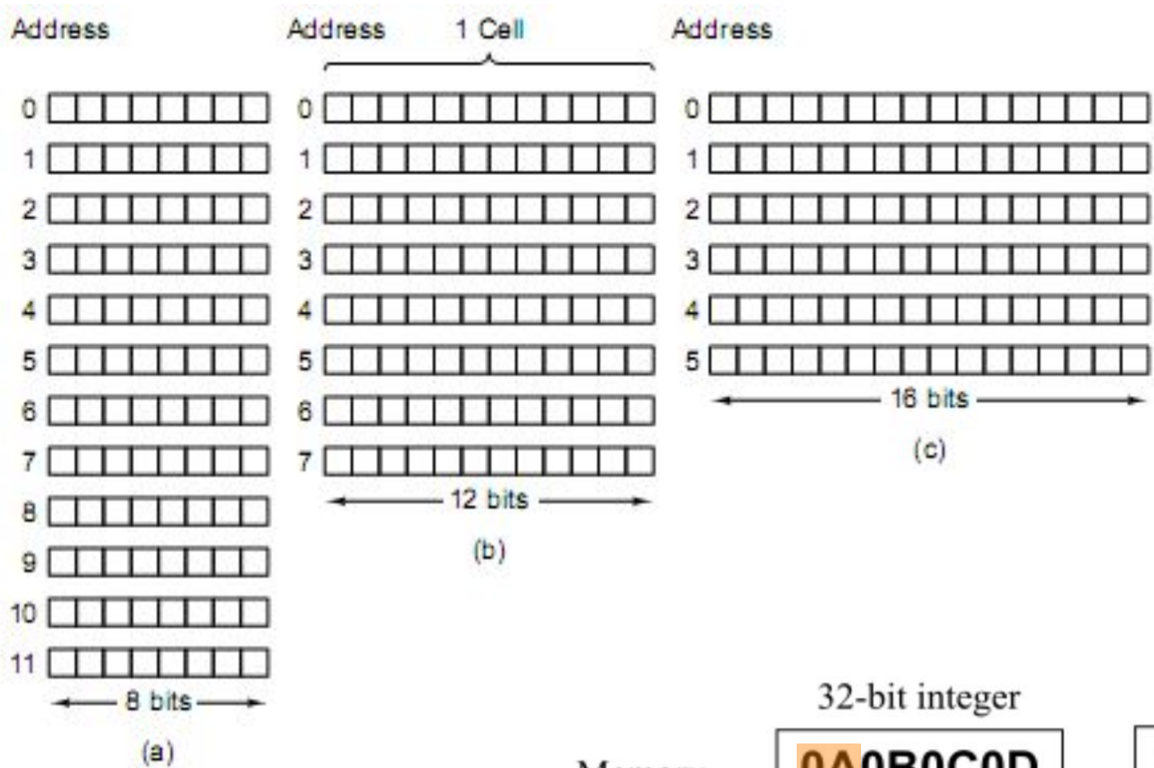
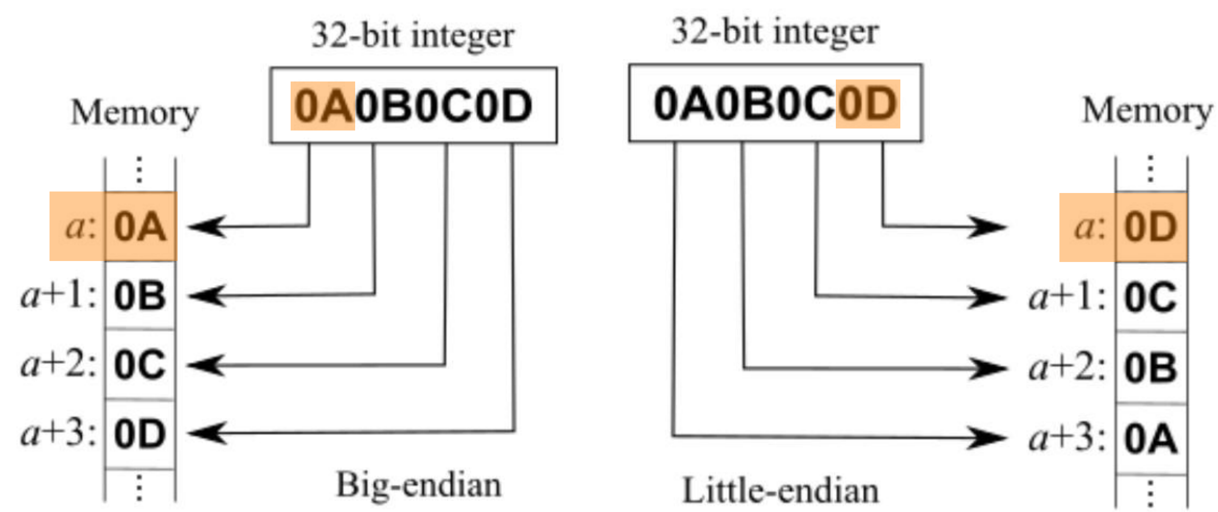


Figure 2-9. Three ways of organizing a 96-bit memory.





Motorola **Intel**

Bit Number	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	0	0	0	0	1	1	0	0	1	0	1	1	0	0	1	0

Fig 4-1. A 16-bit register can hold 16 bits of information

Bit Number	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	0	0	0	0	1	1	0	0	1	0	1	1	0	0	1	0

Fig 4-1. A 16-bit register can hold 16 bits of information

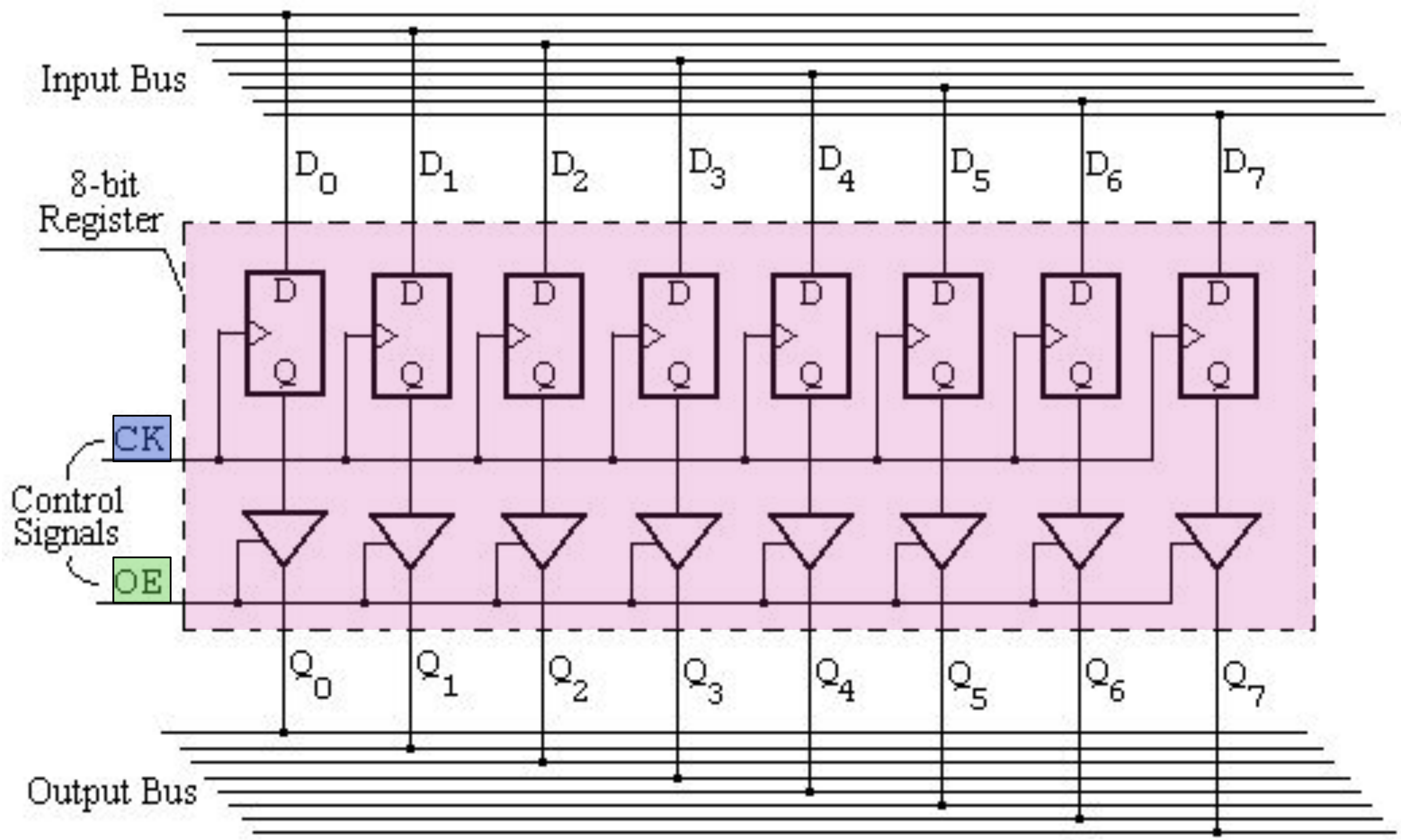


Fig 4-2 (a) Detail of an 8-bit register connected to an input bus and an output bus

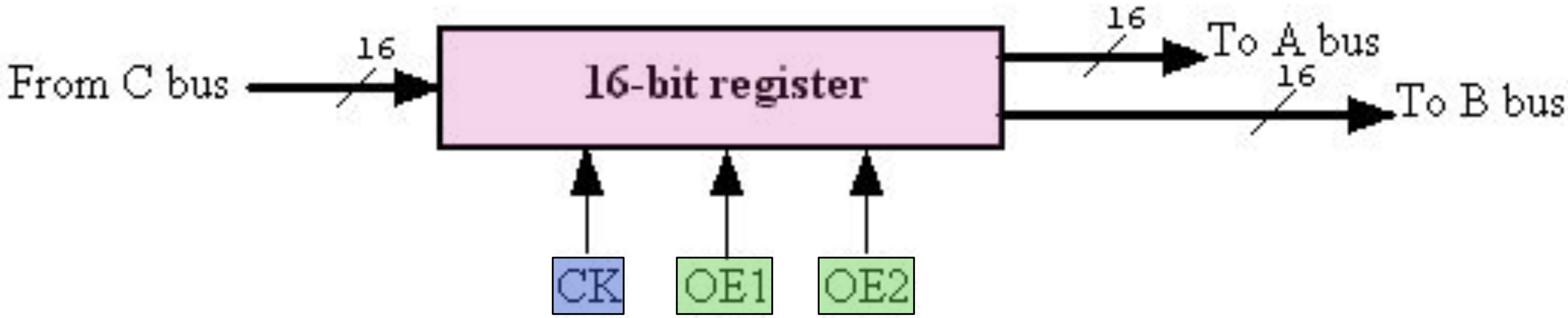


Fig 4-2 (b) Symbolic representation of a 16-bit register with one input bus and two output buses

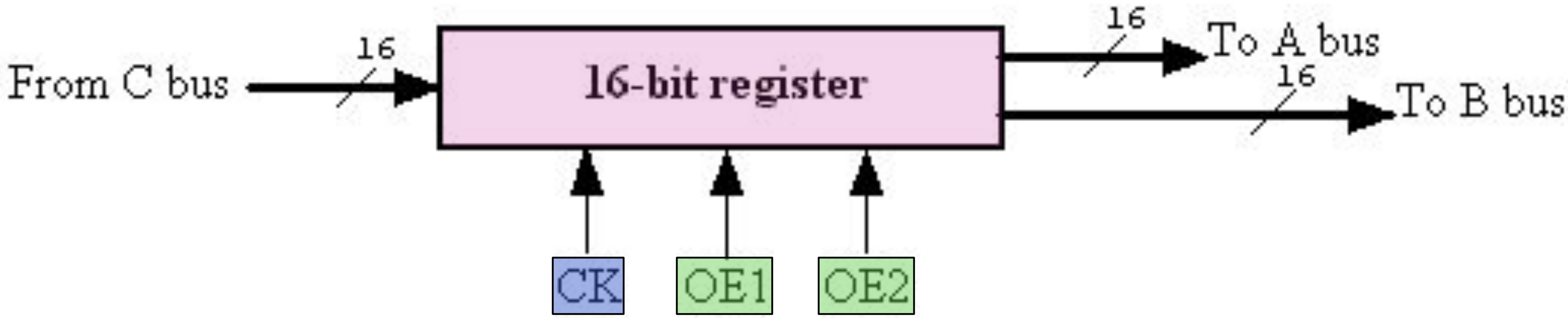
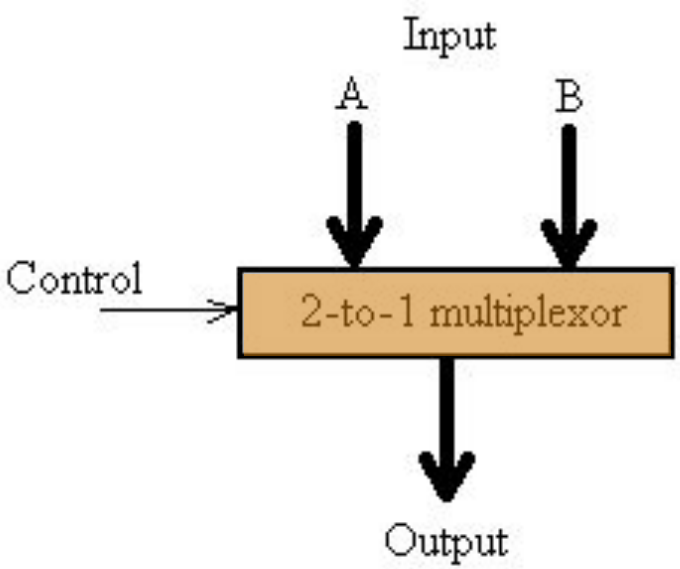
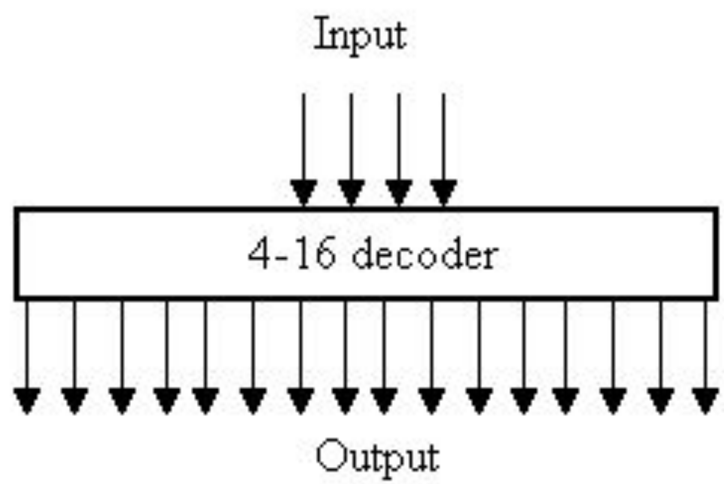


Fig 4-2 (b) Symbolic representation of a 16-bit register with one input bus and two output buses

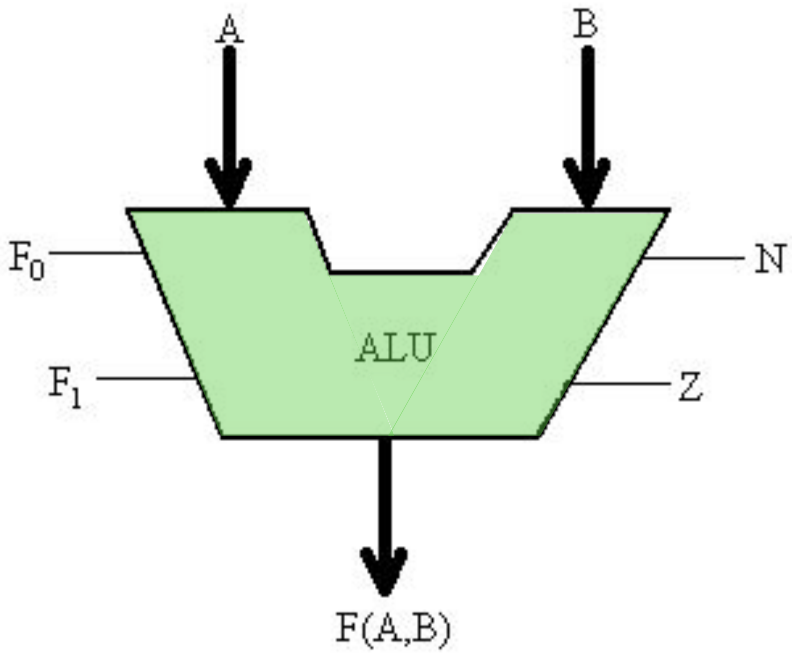


(a)

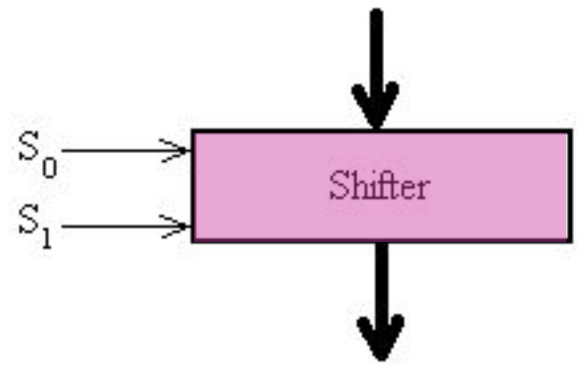


(b)

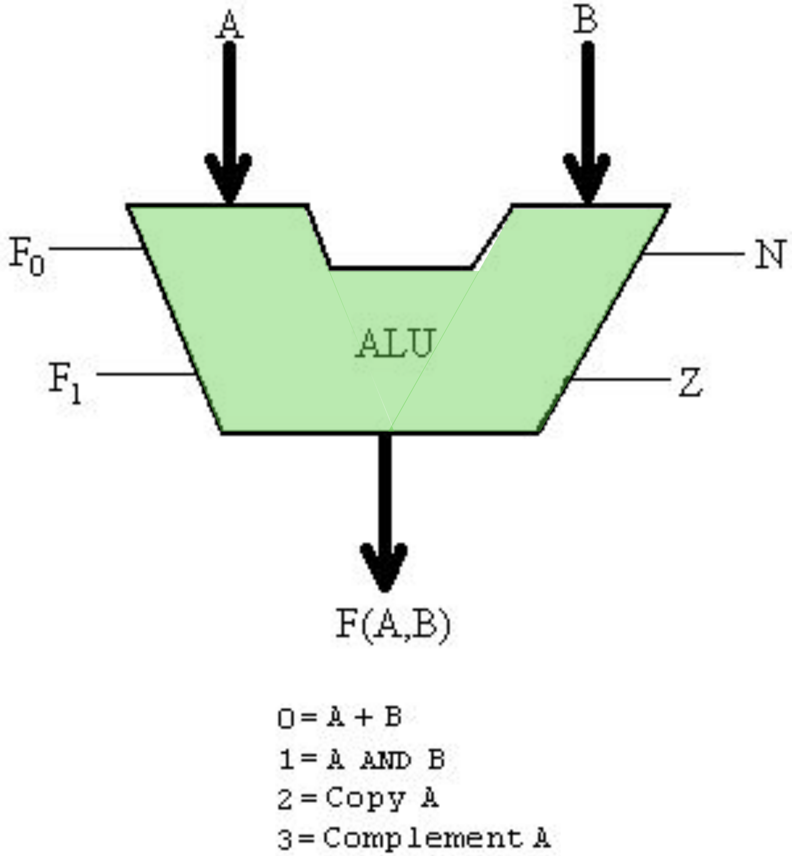
Fig 4-3 (a) A 2-to-1 multiplexor. (b) A 4-to-16 decoder

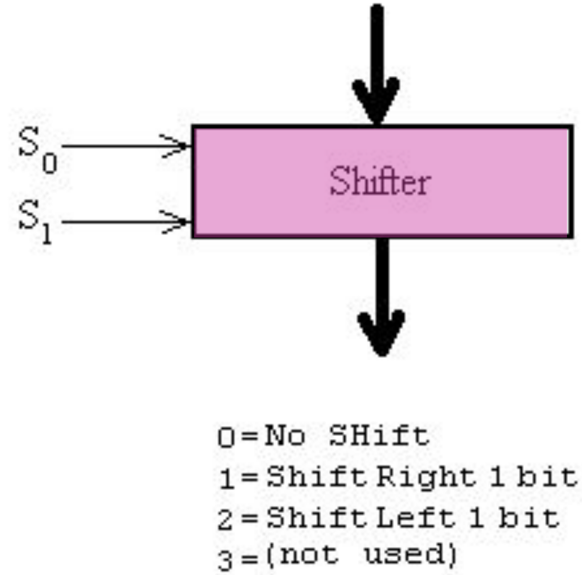


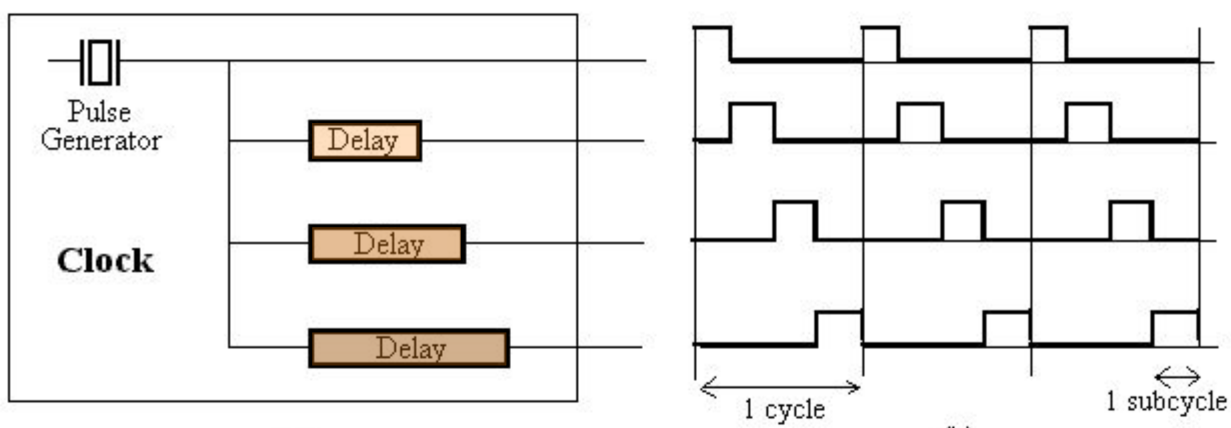
0 = A + B
1 = A AND B
2 = Copy A
3 = Complement A

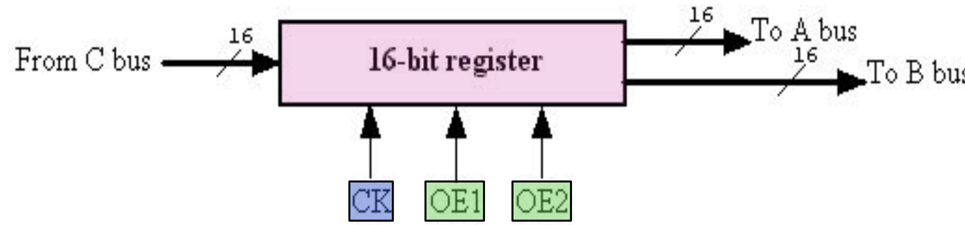


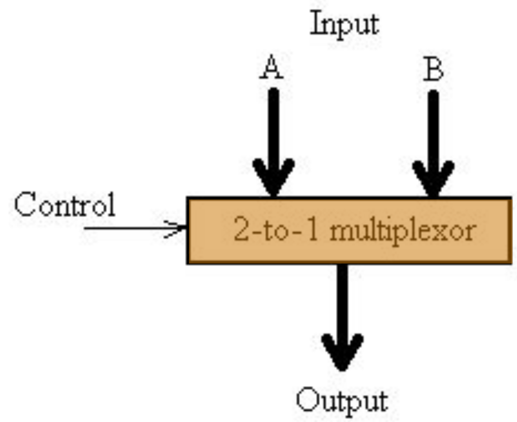
0 = No SHift
1 = Shift Right 1 bit
2 = Shift Left 1 bit
3 = (not used)

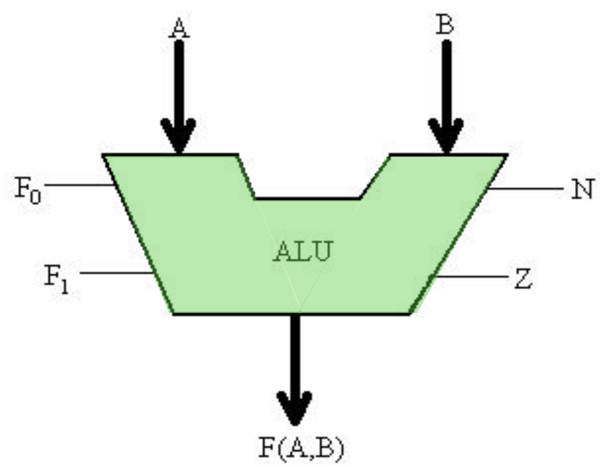

A
B
F0
F1
N
Z
ALU
F(A,B)
0 = A + B
1 = A AND B
2 = Copy A
3 = Complement A

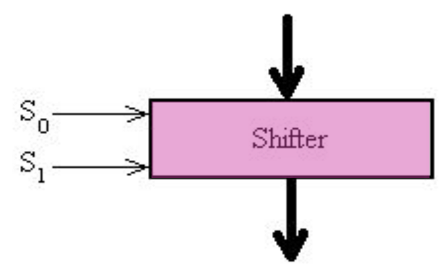

S0
S1
Shifter
0 = No SHift
1 = Shift Right 1 bit
2 = Shift Left 1 bit
3 = (not used)

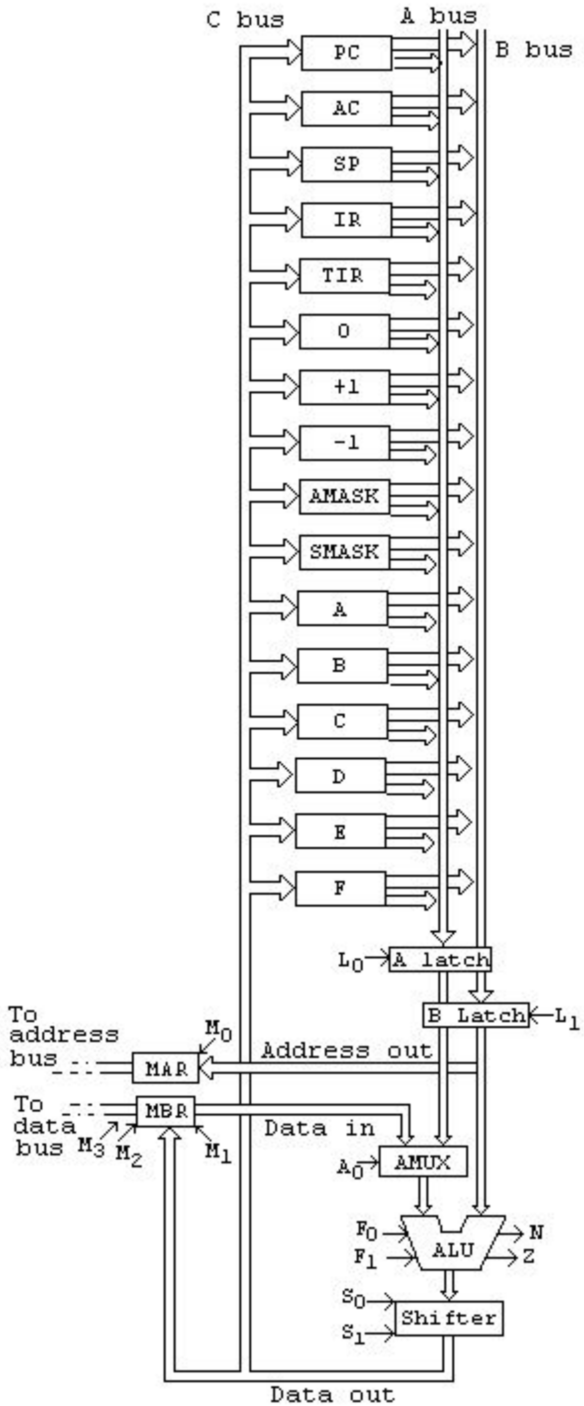

Pulse
Generator
Clock
Delay
Delay
Delay
1 cycle
1 subcycle

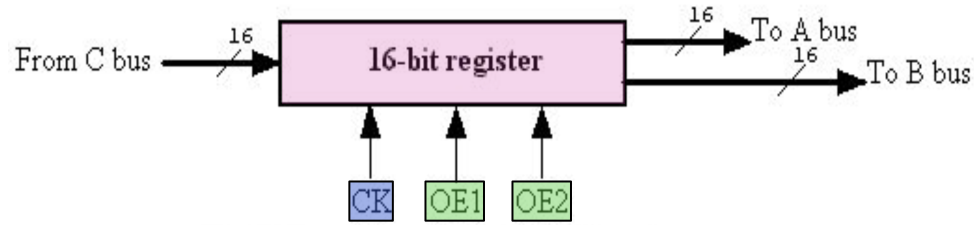

From C bus
16
16-bit register
16
To A bus
16
To B bus
CK
OE1
OE2

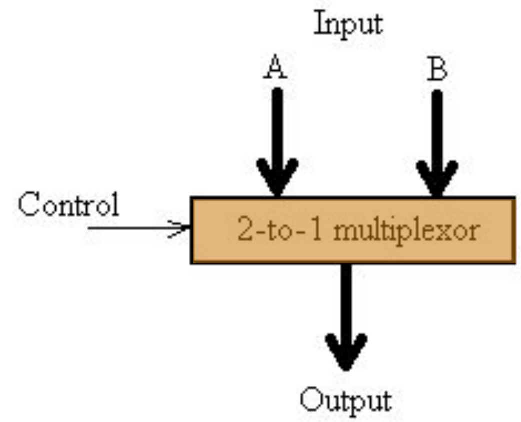

Input
A
B
Control
2-to-1 multiplexor
Output

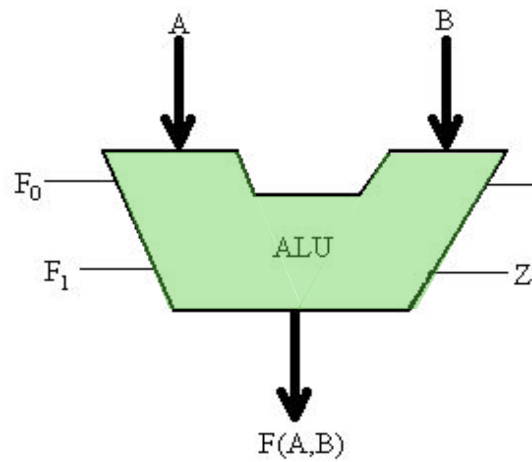

A
B
F0
F1
ALU
N
Z
F(A,B)

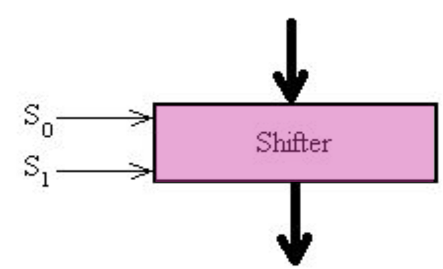

S0
S1
Shifter

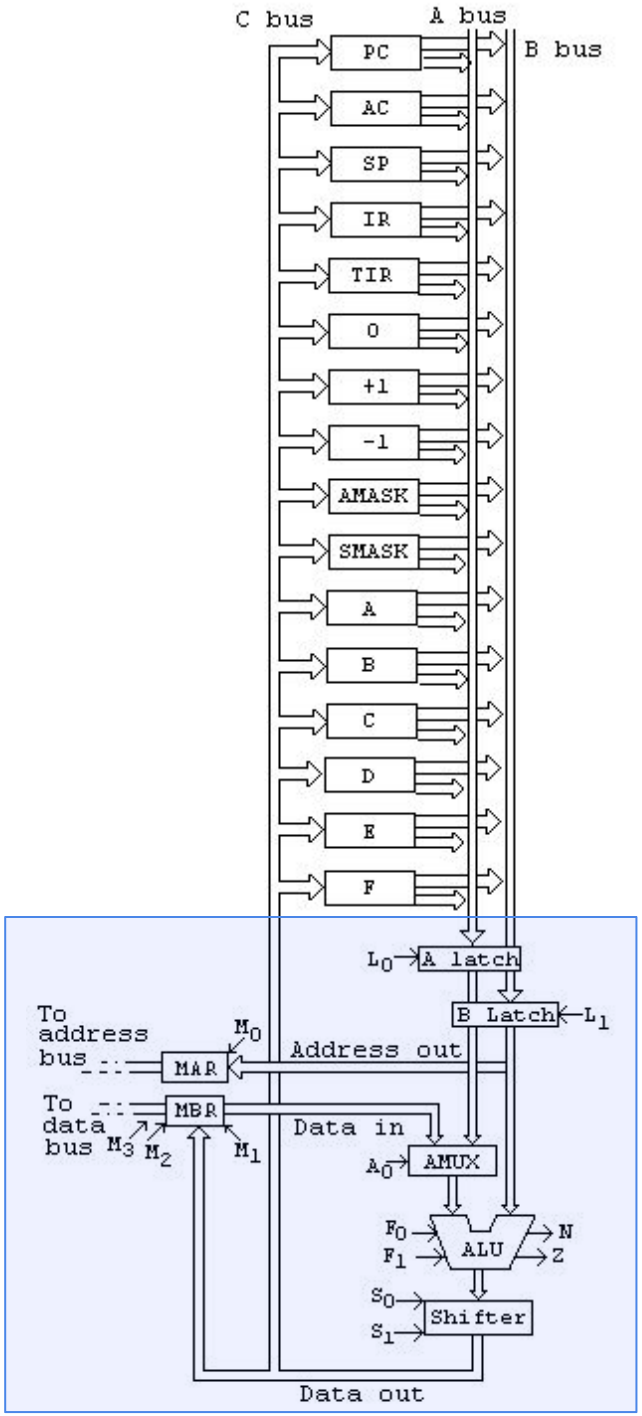

C bus
A bus
B bus
PC
AC
SP
IR
TIR
0
+1
-1
AMASK
SMASK
A
B
C
D
E
F
L_0
A latch
B Latch
L_1
To address bus
M_0
MAR
Address out
To data bus
MBR
M_3
M_2
M_1
Data in
A_0
AMUX
F_0
F_1
ALU
N
Z
S_0
S_1
Shifter
Data out

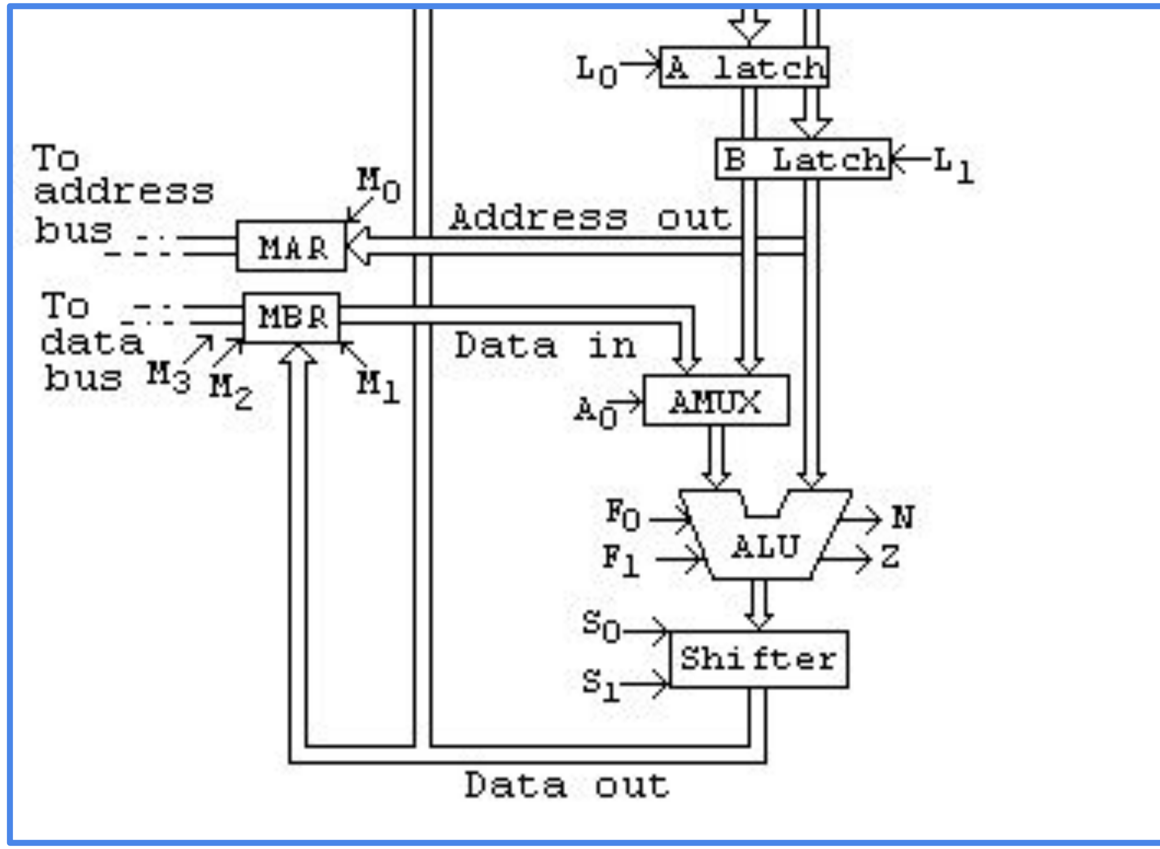

From C bus
16
16-bit register
16
To A bus
16
To B bus
CK
OE1
OE2

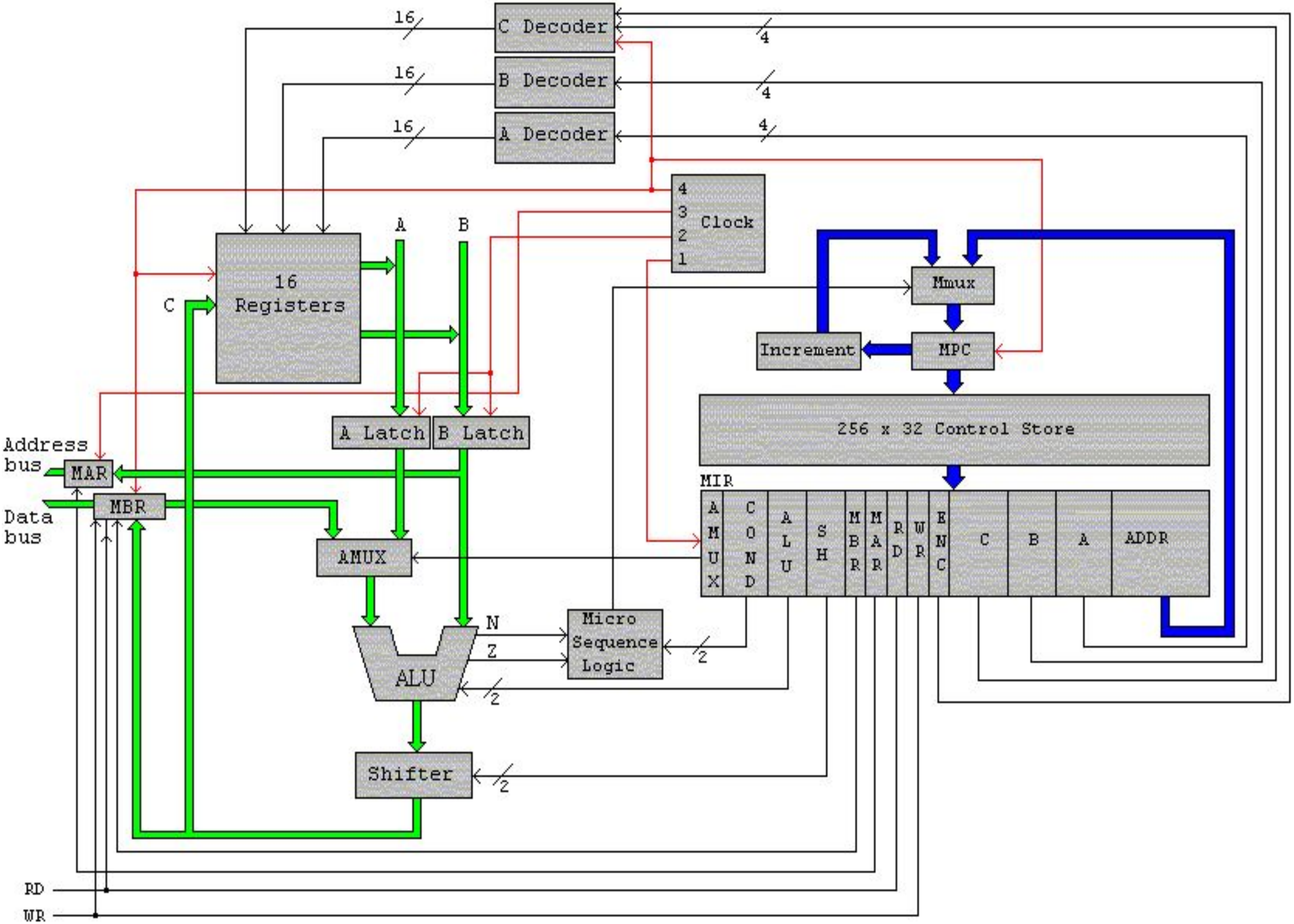

Input
A
B
Control
2-to-1 multiplexor
Output


A
B
F_0
F_1
ALU
N
Z
F(A,B)


S_0
S_1
Shifter


C bus
A bus
B bus
PC
AC
SP
IR
TIR
0
+1
-1
AMASK
SMASK
A
B
C
D
E
F
L_0
A latch
B Latch
L_1
To address bus
M_0
MAR
Address out
To data bus
MBR
M_3
M_2
M_1
Data in
A_0
AMUX
F_0
F_1
ALU
N
Z
S_0
S_1
Shifter
Data out


L_0
A latch
B Latch
L_1
To address bus
M_0
MAR
Address out
To data bus
MBR
M_3
M_2
M_1
Data in
A_0
AMUX
F_0
F_1
ALU
N
Z
S_0
S_1
Shifter
Data out



C Decoder
B Decoder
A Decoder
16
16
16
4
4
4
Clock
4
3
2
1
16
Registers
A
B
C
Mmux
Increment
MPC
256 x 32 Control Store
MIR
AMUX
COND
ALU
SH
MBR
MAR
RD
WR
ENC
C
B
A
ADDR
A Latch
B Latch
Address
bus
MAR
MBR
Data
bus
AMUX
N
Z
Micro
Sequence
Logic
2
ALU
2
Shifter
2
RD
WR

Bits	1	2	2	2	1	1	1	1	1	4	4	4	8
	AMUX	COND	ALU	SH	MBR	MAR	RD	WR	ENC	C	B	A	ADDR

AMUX

0=A Latch
1=MBR

COND

0=No Jump
1=Jump if N=1
2=Jump if Z=1
3=Jump always

ALU

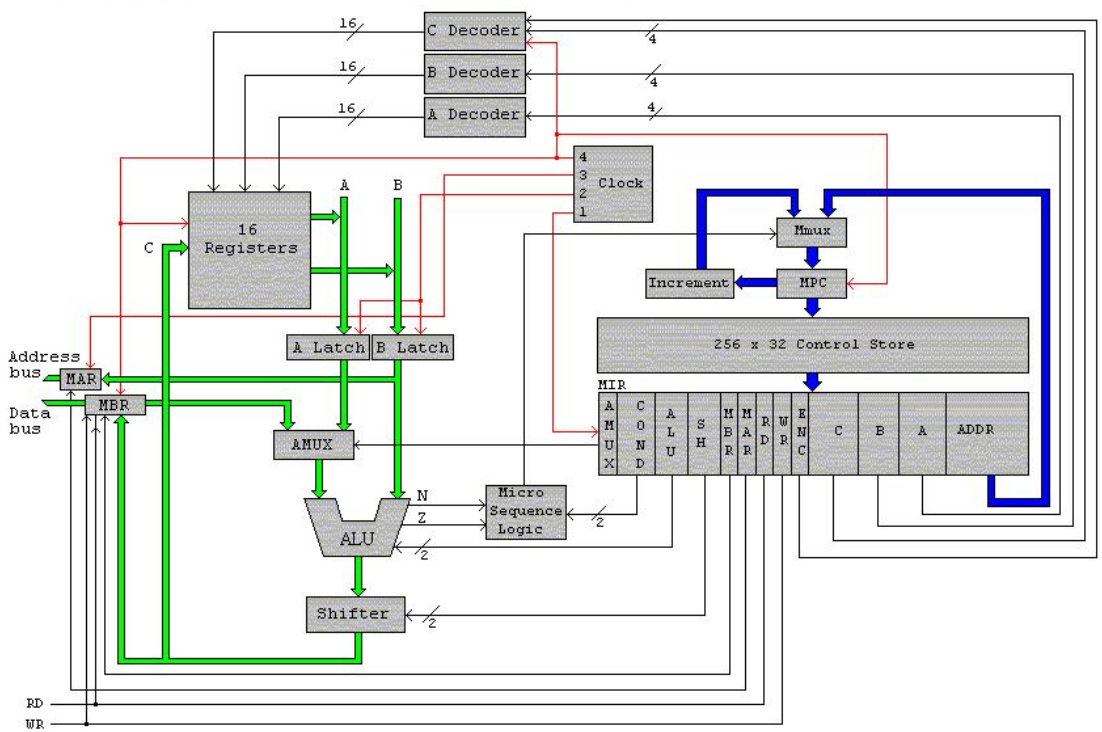
0=A + B
1=A AND B
2=Copy A
3=Complement A

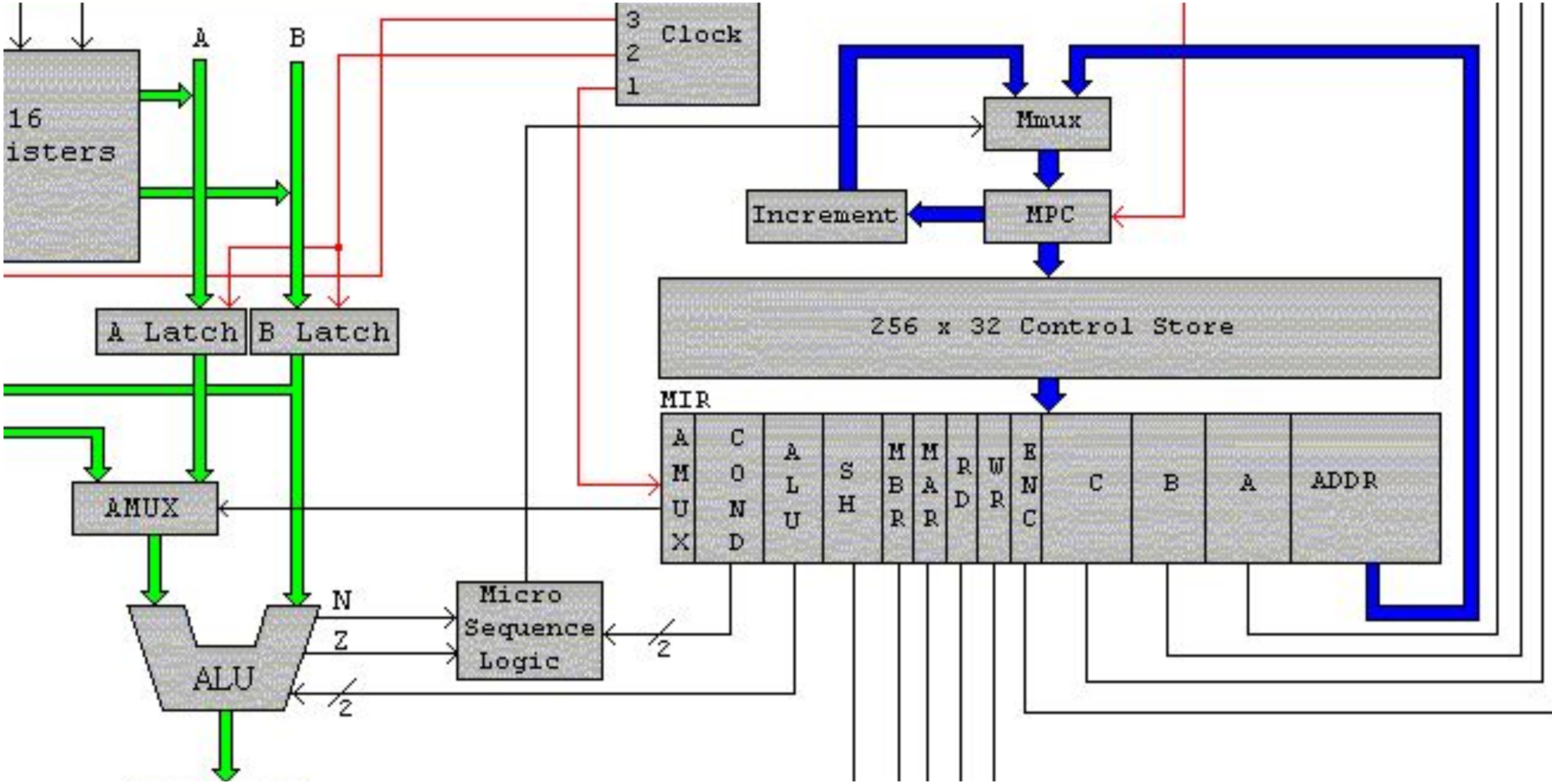
SH

0=No SHift
1=Shift Right 1 bit
2=Shift Left 1 bit
3=(not used)

MBR, MAR, RW, WR, ENC

0 =No
1 =Yes





COND		
0	0	No salta
0	1	Salta si N=1
1	0	Salta si Z=1
1	1	Salta siempre

Máquina Multinivel

Componentes de una arquitectura

Arquitectura de computadoras

Pablo A. Montini
Juan I. Iturriaga
Franco Lanzillotta